



JERUSALEM COLLEGE OF ENGINEERING

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

QUESTION BANK

Regulation	:	R-2023
Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch	:	III / V/ AI&ML
Unit No	:	1
Unit Title	:	INTRODUCTION TO MACHINE LEARNING
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define Machine Learning. Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed. It improves performance through experience.	2	K1	CO1
2	Why Machine Learning Matters? i) Solves complex problems that traditional programming can't. ii) Automate repetitive tasks with accuracy. iii) Handles huge amount of data quickly. iv) Improves and adapts over time with more data.	2	K2	CO1
3	Given vectors $u = [4, 2]$ and $v = [3, 6]$. Identify $u + v$ and $u \cdot v$. $u+v = [4+3, 2+6] = [7, 8]$. $u \cdot v = 4 \cdot 3 + 2 \cdot 6 = 24$	2	K1	CO1
4	List out the objectives of Machine Learning (ML). The objectives of Machine Learning are diverse and depend on the specific application, and they revolve around the following key goals: ✓ Prediction ✓ Classification ✓ Optimization ✓ Pattern Recognition ✓ Automation ✓ Knowledge Discovery ✓ Human-Machine Collaboration	2	K1	CO1



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5	Mention a few applications of Machine Learning. • Disease diagnosis • Fraud detection • Stock price prediction • Product recommendation • Spam email filtering • Customer segmentation • Autonomous vehicles • Demand forecasting	2	K1	CO1															
6	How Machine Learning Impacts Social Platforms? • Personalises Experiences: ML algorithms analyse user behaviour to customise content feeds, predict user preferences, and help businesses design targeted advertising. • Optimises Campaigns: It uncovers key data insights, improves content moderation, and assists marketers in generating and automating social media campaigns.	2	K2	CO1															
7	Outline the differences between supervised learning and unsupervised learning. <table><tr><th>Aspect</th><th>Supervised Learning</th><th>Unsupervised Learning</th></tr><tr><td>Definition</td><td>A ML technique that uses labeled data to train algorithms to predict outcomes and recognize patterns.</td><td>A ML technique that involves training on unlabeled data, allowing the model to discover hidden patterns or structure on its own.</td></tr><tr><td>Data</td><td>Requires input and output data (labeled).</td><td>Requires only input data (unlabeled)</td></tr><tr><td>Goal</td><td>To train the model so that it can predict outcomes on new data.</td><td>To explore the structure of the data to extract meaningful patterns or groupings.</td></tr><tr><td>Types</td><td>Classification and Regression problems.</td><td>Clustering and Association problems.</td></tr></table>	Aspect	Supervised Learning	Unsupervised Learning	Definition	A ML technique that uses labeled data to train algorithms to predict outcomes and recognize patterns.	A ML technique that involves training on unlabeled data, allowing the model to discover hidden patterns or structure on its own.	Data	Requires input and output data (labeled).	Requires only input data (unlabeled)	Goal	To train the model so that it can predict outcomes on new data.	To explore the structure of the data to extract meaningful patterns or groupings.	Types	Classification and Regression problems.	Clustering and Association problems.	2	K1	CO1
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8	Why is deep learning important? Deep learning helps machines learn complex patterns from large data. It is widely used in image, speech, and text processing.	2	K2	CO1
9	Define Null Hypothesis and Alternate Hypothesis. <u>Null Hypothesis (H_0)</u> The null hypothesis is the default assumption or explanation that there is no relation between input features and output variables <u>Alternative Hypothesis (H_1)</u> The alternative hypothesis is a hypothesis that assumes a significant relation between the input data and the output (target value).	2	K1	CO1
10	Define Vapnik-Chervonenkis (VC) Dimension. Give an example. The VC dimension of a hypothesis class is the maximum number of points that can be shattered by that hypothesis class. It measures the capacity or complexity of a learning model. Example: A linear classifier in a 2-dimensional plane has a VC dimension of 3. This means that there exists a set of 3 points that can be shattered by a linear classifier, but no set of 4 points can always be shattered.	2	K1	CO1
11	Why is shattering important in ML? - Shattering means that a hypothesis class can correctly classify every possible labeling of a given set of data points. - Shattering is important because it helps determine the VC dimension, which indicates the capacity and complexity of a learning model. A model with higher capacity can represent more complex patterns, but may also have a greater risk of overfitting.	2	K2	CO1
12	How is Probably Approximately Correct (PAC) learning used in Machine Learning? - PAC learning defines the relationship between sample size, error, and confidence. - It ensures learning is approximately correct with high probability.	2	K2	CO1
13	Illustrate the 2 parameters used in PAC learning. The 2 parameters used in PAC learning are: ϵ and δ . They must have small values. <i>ϵ denotes error</i> $\rightarrow$ Gives the maximum bound on the error in accuracy with which hypotheses h are approximated; hence, accuracy = $1 - \epsilon$. <i>δ denotes probability</i> $\rightarrow$ Gives the probability of failure in achieving this accuracy; hence, confidence = $1 - \delta$.	2	K2	CO1



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14	Compare hypothesis and hypothesis space. Hypothesis: It is the model’s presumption regarding the connection between the input features and the result. Hypothesis Space: The hypothesis space comprises all possible legal hypotheses that a machine learning algorithm can consider.	2	K2	CO1								
15	Define inductive bias. Mention few examples in different ML algorithms. Inductive bias is the assumption made by a model to generalize from training data. Examples: • Linear Regression - Assumes a linear relationship between the input features and the output. • k-Nearest Neighbors (k-NN) - Assumes spatial locality—the belief that data points close to each other in the feature space share the same label or similar outputs.	2	K1	CO1								
16	How generalization is used in ML? • Generalization is the ability of a model to perform well on unseen data. It avoids overfitting and underfitting. • The goal is never to perfectly memorize the past (the training set), but to discover underlying patterns that hold true in the future (the real world).	2	K2	CO1								
17	Define regularization in ML. <i>Regularization</i> is a technique used to improve generalization by adding a penalty to the model's complexity, discouraging the model from fitting the training data too closely, to avoid high variance and overfitting.	2	K1	CO1								
18	Compare overfitting with underfitting. <table><tr><th>Overfitting</th><th>Underfitting</th></tr><tr><td>Model memorises training data and random noise</td><td>Model is too simple to capture underlying patterns</td></tr><tr><td>Excellent on training data but poor on unseen data</td><td>Poor on both training and new data.</td></tr><tr><td>Caused by low bias and high variance.</td><td>Caused by high bias and low variance.</td></tr></table>	Overfitting	Underfitting	Model memorises training data and random noise	Model is too simple to capture underlying patterns	Excellent on training data but poor on unseen data	Poor on both training and new data.	Caused by low bias and high variance.	Caused by high bias and low variance.	2	K2	CO1
Overfitting	Underfitting											
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19	Classify the two key sources of error that affect a model performance on unseen data. Variance - Variance is defined as the variability of model prediction for a given data point which tells us the spread of our data. Bias - Bias is defined as the difference between the prediction of the values by the ML model and the actual / expected values.	2	K2	CO1
20	Define bias-variance trade-off. The bias-variance trade-off is a fundamental concept in machine learning that describes the balances bias and variance to minimize total error. Proper balance improves model performance.	2	K1	CO1

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the various real-world applications of Machine Learning.	16	K2	CO1
2	a) Classify the different types of Machine Learning. b) Summarize the concepts of Linear Algebra used in Machine Learning.	8 8	K2	CO1
3	Describe the role of VC dimension in Machine Learning and explain it with suitable geometric examples.	16	K2	CO1
4	Explain how Probably Approximately Correct (PAC) learning contributes to Machine Learning with suitable details	16	K2	CO1
5	(a) Illustrate the purpose of the bias-variance trade-off in Machine Learning. (b) Describe the concepts of hypothesis and hypothesis space in Machine Learning with suitable examples	8 8	K2	CO1
6	Explain inductive bias and generalization in Machine Learning with suitable examples	16	K2	CO1



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Unit No	:	2
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PART A

Q.No.	Questions	Marks	BTL	CO																								
1	Define Supervised Learning. Give its types. Supervised learning is a type of machine learning where a model is trained on a labeled dataset. In this context, "labeled" means that each training example is paired with an output label. Types of Supervised Learning: 1) Classification 2) Regression	2	K1	CO2																								
2	Give an example of dataset that can be modeled using regression. <table><tr><th>Quantity Sold</th><th>Price</th><th>Amount for Advertising</th></tr><tr><td>8500</td><td>\$2</td><td>\$2,800</td></tr><tr><td>4700</td><td>\$5</td><td>\$200</td></tr><tr><td>5800</td><td>\$3</td><td>\$400</td></tr><tr><td>7400</td><td>\$2</td><td>\$500</td></tr><tr><td>6200</td><td>\$5</td><td>\$3,200</td></tr><tr><td>7300</td><td>\$3</td><td>\$1,800</td></tr><tr><td>5600</td><td>\$4</td><td>\$900</td></tr></table> Here the amount to be spent for ‘Advertising’ can be predicted using ‘Quantity Sold’ and ‘Price’ for an item.	Quantity Sold	Price	Amount for Advertising	8500	\$2	\$2,800	4700	\$5	\$200	5800	\$3	\$400	7400	\$2	\$500	6200	\$5	\$3,200	7300	\$3	\$1,800	5600	\$4	\$900	2	K1	CO2
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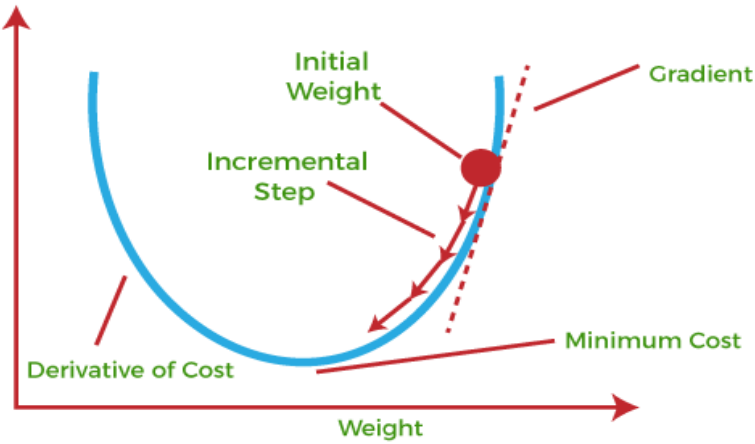
3	Mention few examples and applications of supervised learning algorithms. <i>Examples of supervised learning algorithms</i> • Linear Regression • Logistic Regression <i>Applications of supervised learning algorithms</i> • Natural Language Processing (NLP) • Fraud Detection	2	K1	CO2
4	Define regression algorithms. Regression algorithms are used when output is continuous. They model relationships between variables.	2	K1	CO2
5	List out the types of Linear Regression. • Simple Linear Regression • Multiple Linear Regression • Lasso Regression (L1 Regularization) • Ridge Regression (L2 Regularization) • Polynomial Regression	2	K1	CO2
6	Outline the Simple Linear Regression equation. $Y = \beta_0 + \beta_1 X + \epsilon$ • Y represents the output or target or dependent variable. • X represents the input or independent variable • β_0 represents unknown constant called the intercept • β_1 represents unknown constant called coefficient (slope) • ϵ (Epsilon) is the error term.	2	K2	CO2
7	Illustrate two applications of Simple Linear Regression with examples. Used in predicting salary, crop yield, and student marks. Helps forecast continuous outcomes. Example: (i) Crop Yield Prediction: ✓ A farmer predicts crop yield based on the amount of fertilizer used. ✓ Increasing fertilizer from 50 kg to 70 kg may increase the expected yield. (ii) Student Marks Prediction: ✓ A teacher predicts a student's exam marks based on the number of study hours. ✓ A student studying 8 hours is likely to score higher than one studying 4 hours.	2	K2	CO2



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8	Briefly describe the equation of Multiple Linear Regression. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$ • $Y \rightarrow$ the output or target or dependent variable. • $x_1, x_2, \dots, x_n \rightarrow$ the input or independent variables • $\beta_0 \rightarrow$ the intercept $\rightarrow$ the value of y (response variable) when the values of all the predictor variables (X_s) are zero • $\beta_1, \beta_2, \dots, \beta_n \rightarrow$ the coefficients of the independent variables. • $\epsilon \rightarrow$ the error term.	2	K2	CO2
9	Define Least Squares Error Method. • The least squares method is a mathematical approach used in regression analysis to find the best-fitting line or curve to a set of data points. • The "best-fitting" line or curve is the one that minimizes the sum of the squared differences between the observed values (actual data points) and the values predicted by the model. Sum of squares of errors (SSE) = $\sum_{i=1}^n (y_i - \bar{y}_i)^2$	2	K1	CO2
10	Outline the Gradient Descent optimization algorithm. • Gradient Descent is an optimization algorithm used to minimize the loss function in machine learning models. • It iteratively updates model parameters in the direction of the negative gradient to reach the minimum error. 	2	K2	CO2
11	Discuss the advantages of Batch Gradient Descent. • It produces less noise in comparison to other gradient descent. • It produces stable gradient descent convergence. • It is computationally efficient as all resources are used for all training samples.	2	K2	CO2



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12	Why is SVM important in ML? - Support Vector Machine (SVM) is a powerful supervised learning algorithm used for classification and regression. - It is effective in high-dimensional spaces and provides accurate results with less overfitting.	2	K2	CO2										
13	Compare Stochastic Gradient Descent (SGD) and Mini-Batch Gradient Descent. <table><tr><th>Stochastic Gradient Descent</th><th>Mini-Batch Gradient Descent</th></tr><tr><td>Single observation is used for each parameter update.</td><td>A subset (mini-batch) of the training data is used for each update.</td></tr><tr><td>The cost (loss) function shows high variations and is noisy.</td><td>The cost (loss) function is smoother compared to SGD.</td></tr><tr><td>Computation time is more due to frequent parameter updates.</td><td>Computation time is less than SGD.</td></tr><tr><td>Less memory is required since one sample is processed at a time.</td><td>Computational cost is less than Batch Gradient Descent while maintaining good performance.</td></tr></table>	Stochastic Gradient Descent	Mini-Batch Gradient Descent	Single observation is used for each parameter update.	A subset (mini-batch) of the training data is used for each update.	The cost (loss) function shows high variations and is noisy.	The cost (loss) function is smoother compared to SGD.	Computation time is more due to frequent parameter updates.	Computation time is less than SGD.	Less memory is required since one sample is processed at a time.	Computational cost is less than Batch Gradient Descent while maintaining good performance.	2	K2	CO2
Stochastic Gradient Descent	Mini-Batch Gradient Descent													
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Less memory is required since one sample is processed at a time.	Computational cost is less than Batch Gradient Descent while maintaining good performance.													
14	Illustrate the use of Discriminant Function. - A discriminant function is used in pattern classification to assign an input to one of the classes by evaluating its features. - It helps partition the feature space into different classes.	2	K2	CO2										
15	Define Binary Classifiers. A binary classifier is a model that classifies data into two classes, such as 0 and 1 or positive and negative. It is widely used in applications like spam detection and medical diagnosis.	2	K1	CO2										
16	Describe the use of Linear Discriminant Analysis. - LDA is a dimensionality reduction technique used to improve class separability. - It projects data onto a lower-dimensional space while maximizing the distance between classes. - In real-time, it is used for Face recognition and medical diagnosis.	2	K2	CO2										



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17	Outline the Perceptron learning algorithm. The perceptron is a simple neural network used for binary classification. It computes a weighted sum of inputs, adds bias, and applies an activation function to produce output.	2	K2	CO2
18	List any four discriminative models. • Support Vector Machine • Logistic Regression • Decision Trees • Random Forest	2	K1	CO2
19	Describe two major generative models There are two main types of generative models: Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs). • GANs: - consists of two neural networks, a generator and a discriminator, that compete with each other in order to generate realistic data. • VAEs: - consist of an encoder and a decoder, which work together to compress data into latent variables and then generate new data from these latent variables.	2	K2	CO2
20	Mention any four applications of Naïve Bayes Classifier. • Spam email detection • Text/document classification • Sentiment analysis • Medical diagnosis	2	K1	CO2



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PART B

Q.No.	Questions	Marks	BTL	CO																																	
1	The following data represents the number of hours 12 different students watched television during the weekend and the scores of each student who took a test the following Monday. A) Find the equation of the regression line B) Use the equation to find the expected test score for a student who watches 9 hours of TV. <table><tr><td>Hours, x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>3</td><td>5</td><td>5</td><td>5</td><td>6</td><td>7</td><td>7</td><td>10</td></tr><tr><td>Test score, y</td><td>96</td><td>85</td><td>82</td><td>74</td><td>95</td><td>68</td><td>76</td><td>84</td><td>58</td><td>65</td><td>75</td><td>50</td></tr></table>	Hours, x	0	1	2	3	3	5	5	5	6	7	7	10	Test score, y	96	85	82	74	95	68	76	84	58	65	75	50	16	K3	CO2							
Hours, x	0	1	2	3	3	5	5	5	6	7	7	10																									
Test score, y	96	85	82	74	95	68	76	84	58	65	75	50																									
2	A real estate company wants to predict the selling price (Y in Lakh) of a house based on its size (X_1 in 1000 sq.ft) and number of bedrooms (X_2) using Multiple Linear Regression. The total no. of houses is 10. <table><tr><td>X_1</td><td>1.5</td><td>1.8</td><td>2.4</td><td>1.2</td><td>3.0</td><td>1.6</td><td>2.1</td><td>2.7</td><td>1.4</td><td>2.2</td></tr><tr><td>X_2</td><td>3</td><td>4</td><td>3</td><td>2</td><td>5</td><td>3</td><td>4</td><td>4</td><td>2</td><td>3</td></tr><tr><td>Y</td><td>2.5</td><td>3.0</td><td>3.8</td><td>2.0</td><td>4.8</td><td>2.7</td><td>3.4</td><td>4.3</td><td>2.2</td><td>3.6</td></tr></table> Use the model of the form: $Y=b_0+b_1X_1+b_2X_2$. Where, b_0 is the intercept and b_1, b_2 are the regression coefficients. Predict the house price (Y) for a house with, House Size (X_1) = 2.5 (1000 sq. ft.) and Number of Bedrooms (X_2) = 3	X_1	1.5	1.8	2.4	1.2	3.0	1.6	2.1	2.7	1.4	2.2	X_2	3	4	3	2	5	3	4	4	2	3	Y	2.5	3.0	3.8	2.0	4.8	2.7	3.4	4.3	2.2	3.6	16	K3	CO2
X_1	1.5	1.8	2.4	1.2	3.0	1.6	2.1	2.7	1.4	2.2																											
X_2	3	4	3	2	5	3	4	4	2	3																											
Y	2.5	3.0	3.8	2.0	4.8	2.7	3.4	4.3	2.2	3.6																											
3	a) Explain in detail about Bayesian Linear Regression technique. b) Briefly explain the Least Square Method.	8 8	K2	CO2																																	
4	a) Describe in detail about how Random Forest classifier. b) Compare and contrast Probabilistic Generative Model and Probabilistic Discriminative Model.	8 8	K2	CO2																																	
5	Describe how a margin classifier works in separating data points in SVM.	16	K2	CO2																																	
6	Compute a decision tree, which predicts whether to Play or Not Play tennis, using the following training data. Also, classify the given tuple: $X = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$	16	K3	CO2																																	



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Regulation	:	R-2023
Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch		III/ V/AI&ML
Unit No	:	3
Unit Title	:	ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING
No. of Hours	:	9
Weightage in Exam (%)	:	20

QUESTION BANK

PART A

Q.No.	Questions	Marks	BTL	CO
1	State the function of ensemble techniques. It is a powerful machine learning technique that involves combining the predictions from multiple models to improve overall performance and to produce one optimal predictive model.	2	K1	CO3
2	List few ensemble techniques. • Voting • Averaging • Weighted averaging • Stacking • Bagging • Boosting	2	K1	CO3
3	Mention the different methods of generating diverse learners. Diversity among learners is achieved using different algorithms, datasets, or hyperparameters. This improves the robustness and performance of ensemble models.	2	K1	CO3
4	Illustrate the use of Bootstrap Sampling. • Bootstrap sampling is a technique in which multiple subsets of data are created by sampling with replacement from the original dataset.	2	K2	CO3



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	It is mainly used to train multiple models in ensemble methods.															
5	Discuss about the algorithms used in Stacking. Algorithms used in stacking include Generalized Stacking, where different base models are trained and a meta-model combines their predictions. Stacking with Cross-Validation uses validation predictions from base models to train the meta-model, while Multi-Layer Stacking uses multiple levels of models before final prediction.	2	K2	CO3												
6	How does Voting in machine learning work? Voting is an ensemble method where multiple models make predictions and the final output is obtained by majority voting (classification) or averaging (regression). It is simple and effective.	2	K2	CO3												
7	Interpret how bagging improves the stability and accuracy of a model in machine learning. Bagging (Bootstrap Aggregating) involves training multiple models on different random subsets of data. It reduces variance and improves model stability.	2	K2	CO3												
8	Outline the concept of boosting technique. Boosting is an ensemble technique where models are trained sequentially. Each model focuses on correcting the errors of the previous model, improving overall performance.	2	K2	CO3												
9	How is the Stacking technique used in ensemble learning? Stacking combines multiple base models by using their outputs as inputs to a meta-model. This approach improves prediction accuracy by leveraging strengths of different models.	2	K2	CO3												
10	<table><tr><td colspan="3">Compare Bagging and Boosting.</td></tr><tr><td>Aspect</td><td>Bagging</td><td>Boosting</td></tr><tr><td>Model Training</td><td>Multiple models are trained independently and in parallel on different random subsets of data.</td><td>Models are trained sequentially, with each new model focusing on correcting the errors of the previous one.</td></tr><tr><td>Objective/ Weighting</td><td>Reduces variance by averaging results; each model gets equal</td><td>Reduces both bias and variance; models are weighted based on their accuracy.</td></tr></table>	Compare Bagging and Boosting.			Aspect	Bagging	Boosting	Model Training	Multiple models are trained independently and in parallel on different random subsets of data.	Models are trained sequentially, with each new model focusing on correcting the errors of the previous one.	Objective/ Weighting	Reduces variance by averaging results; each model gets equal	Reduces both bias and variance; models are weighted based on their accuracy.	2	K2	CO3
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		weight in the final decision.				
11	State Unsupervised Learning with one suitable example. Unsupervised learning is a type of machine learning that works with unlabeled data. It is used to discover hidden patterns, structures, or relationships in the data. Example: Customer segmentation using clustering.			2	K1	CO3
12	Mention few examples of Unsupervised Learning algorithms. • K-means clustering • Principal Component Analysis • Hierarchical clustering			2	K1	CO3
13	Define clustering. Specify how it is used in K-means algorithm. Clustering is a technique used to group similar data points together. K-means is an algorithm that divides data into K clusters based on similarity.			2	K1	CO3
14	Define Instance Based Learning. Instance-based learning stores training data and makes predictions by comparing new data with stored instances. A common example is the K-Nearest Neighbors (KNN) algorithm.			2	K1	CO3
15	How does K-means algorithm differ from KNN algorithm? • KNN is a supervised learning algorithm mainly used for classification problems, whereas K-Means (aka K-means clustering) is an unsupervised learning algorithm. • K in K-Means refers to the number of clusters, whereas K in KNN is the number of nearest neighbors (based on the chosen distance metric).			2	K2	CO3
16	Outline the benefits of K-NN algorithm. • Simplicity and ease of implementation • No explicit training phase is required. • Used for both classification and regression tasks • Non-parametric - does not assume any specific data distribution • Supports multi-class classification • Inherent outlier detection			2	K2	CO3
17	Define Gaussian Mixture Models (GMM).			2	K1	CO3



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	GMM is a probabilistic model that represents data as a mixture of multiple Gaussian distributions. It is widely used for clustering tasks.					
18	Compare Gaussian Mixture Model with K-Means Algorithm.		2	K2	CO3	
	Feature	Gaussian Mixture Model (GMM)				K-Means Clustering
	Clustering Approach	Probabilistic (soft clustering)				Deterministic (hard clustering)
	Cluster Shape	Can handle ellipsoidal clusters with varying sizes				Assumes spherical, equally-sized clusters
	Assignment	Data points have probabilities of belonging to clusters				Data points are assigned to a single cluster
	Flexibility	Models complex data distributions				Works best with well-separated, simple data
	Parameters	Incorporates means, covariances, and mixing coefficients				Only considers cluster centroids
19	How does Expectation Maximization algorithm used in ML? EM is an algorithm used to estimate parameters of probabilistic models with hidden variables. It iteratively improves model parameters.		2	K2	CO3	
20	How does the Expectation-Maximization (EM) algorithm work during implementation? The two steps of the EM algorithm are: <i>E-step</i> : perform probabilistic assignments of each data point to some class based on the current hypothesis h for the distributional class parameters; <i>M-step</i> : update the hypothesis h for the distributional class parameters based on the new data assignments.		2	K2	CO3	



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PART B

Q.No.	Questions	Marks	BTL	CO																																													
1	i) Explain the purpose of Voting in Machine Learning with example.	8	K2	CO3																																													
	ii) Explain the working of Stacking as an Ensemble Learning technique.	8	K2	CO3																																													
2	Discuss about Bagging and Boosting in Ensemble Learning.	16	K2	CO3																																													
3	Apply various Ensemble Learning Techniques to a given classification problem and describe their outcomes.	16	K3	CO3																																													
4	Using the K-means algorithm, cluster the following eight points (x, y) into three clusters: $A_1 = (2, 10)$, $A_2 = (2, 5)$, $A_3 = (8, 4)$, $B_1 = (5, 8)$, $B_2 = (7, 5)$, $B_3 = (6, 4)$ Initial Cluster Point: $C_1 = (1, 2)$ and $C_2 = (4, 9)$ Use Euclidean Distance Function and show (i) Three cluster centres after the first-round execution (ii) The final three clusters.	16	K3	CO3																																													
5	Explain in detail the Expectation-Maximization (EM) algorithm in the context of Gaussian Mixture Models.	16	K2	CO3																																													
6	Consider the given training dataset and use the KNN algorithm to determine the class of the test instance $X = (7.6, 60, 8)$ with $K = 3$. <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>S.No.</th><th>CGPA</th><th>Assessment</th><th>Project Submitted</th><th>Result (Class)</th></tr> </thead> <tbody> <tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>Pass</td></tr> <tr><td>2</td><td>8</td><td>80</td><td>7</td><td>Pass</td></tr> <tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>Pass</td></tr> <tr><td>4</td><td>6</td><td>45</td><td>5</td><td>Fail</td></tr> <tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>Fail</td></tr> <tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>Pass</td></tr> <tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>Fail</td></tr> <tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>Pass</td></tr> </tbody> </table>	S.No.	CGPA	Assessment	Project Submitted	Result (Class)	1	9.2	85	8	Pass	2	8	80	7	Pass	3	8.5	81	8	Pass	4	6	45	5	Fail	5	6.5	50	4	Fail	6	8.2	72	7	Pass	7	5.8	38	5	Fail	8	8.9	91	9	Pass	16	K3	CO3
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